

Companion to Bird & de Moor, Algebra of Programming

§10. Greedy Algorithms

§10.1. Theory

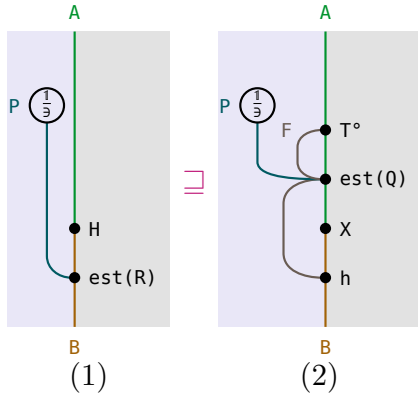
(10.1a)

h, T, R, H, M as in $M \triangleq \frac{H}{\exists} \text{ est}(R)$; additionally Q a **connected** preorder on the sets $\frac{T^\circ}{\exists}$ returns, so that $\frac{T^\circ}{\exists} \text{ est}(Q)$ is entire.

$\text{Map}(h) \wedge h \text{ Pres } R \wedge R \in R \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow (\mu X : \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h) \sqsubseteq \frac{H}{\exists} \text{ est}(R)$ Theorem 10.1

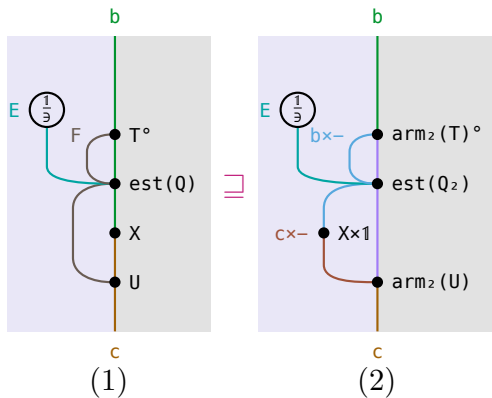
the same optimum reached by keeping ONE decomposition at each step, so that no set is ever carried and the recursion runs on values alone

$H \triangleq (T)^\circ (h) \triangleq H, Q$



1	$\frac{H}{\exists} \text{ est}(R)$	
2	$\sqsubseteq \frac{T^\circ}{\exists} \text{ est}(Q) F(X) h$	

the body at T a coproduct of two arms

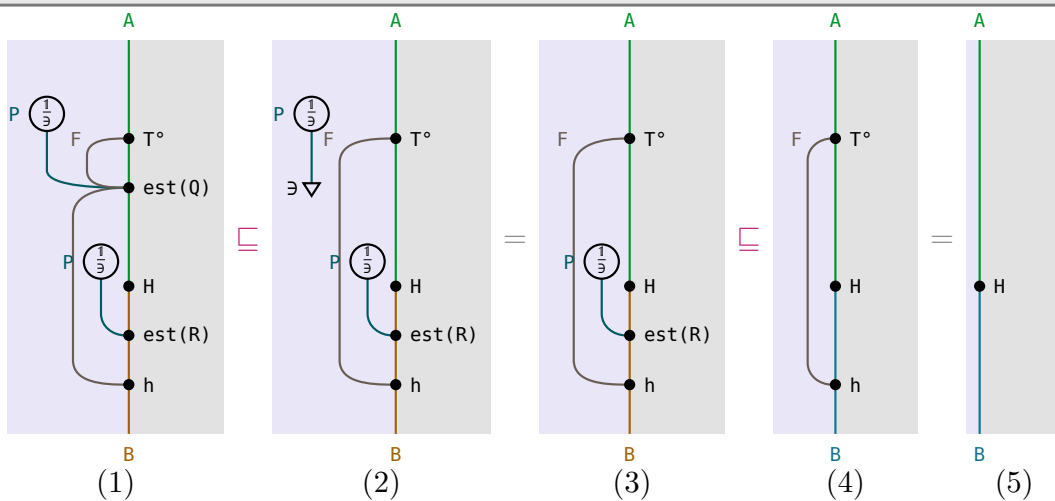


1	$\frac{T^\circ}{\exists} \text{ est}(Q) F(X) U$	Proposition 10.1, $\forall_2 \forall_1^\circ = \emptyset$
2	$\sqsubseteq \frac{\text{arm}_2(T)^\circ}{\exists} \text{ est}(Q_2) (X \times 1) \text{ arm}_2(U)$	

$\frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h \sqsubseteq H$

(10.1b)

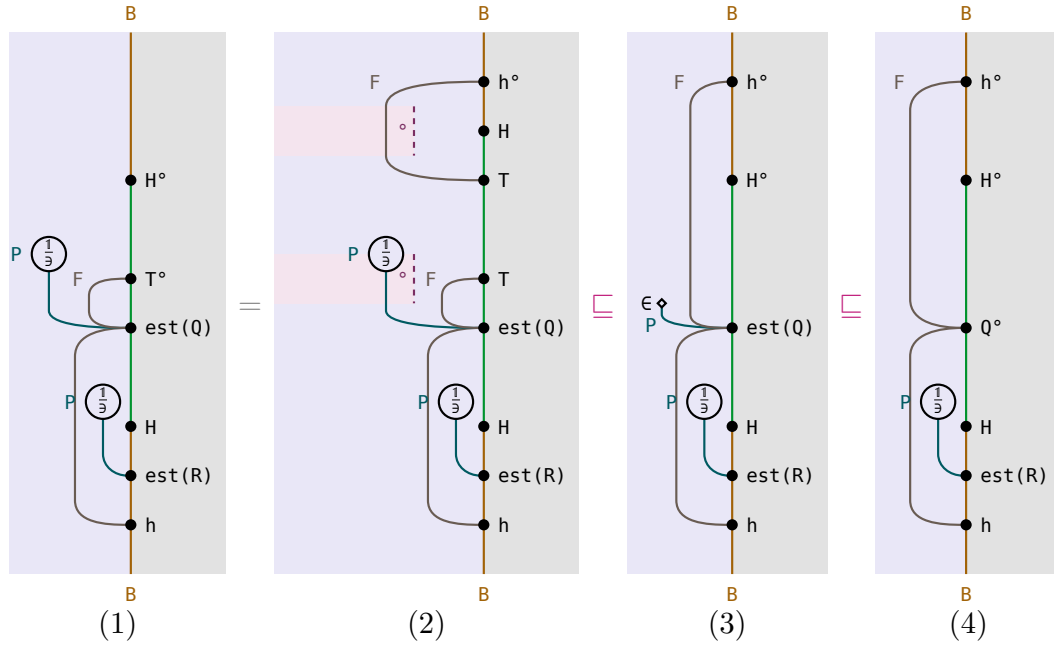
taking the input apart in one Q -extreme way, solving the parts by M and assembling by h returns only what H returns



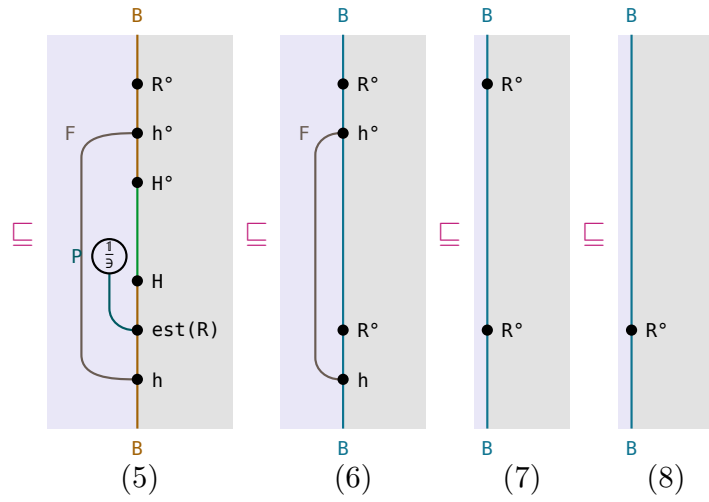
1	$\frac{T^\circ}{\exists} \text{ est}(Q) F(\frac{H}{\exists} \text{ est}(R)) h$	$\text{est}(Q) \sqsubseteq \exists$
2	$\sqsubseteq \frac{T^\circ}{\exists} \exists F(\frac{H}{\exists} \text{ est}(R)) h$	$\frac{R}{\exists} \exists = R$
3	$= T^\circ F(\frac{H}{\exists} \text{ est}(R)) h$	$\frac{\text{est}(R)}{\exists} \sqsubseteq S$
4	$\sqsubseteq T^\circ F(H) h$	$T^\circ F(H) h = H$
5	$= H$	

$$\text{Map}(h) \wedge h \text{ Pres } R \wedge R R \subseteq R \wedge \text{ThinCondition}(T, h, R, Q) \Rightarrow H^\circ \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h \subseteq R^\circ \quad (10.1c)$$

H° followed by the greedy body at M is $\exists R^\circ$: an answer of the body is never worse than an answer of H to the same input



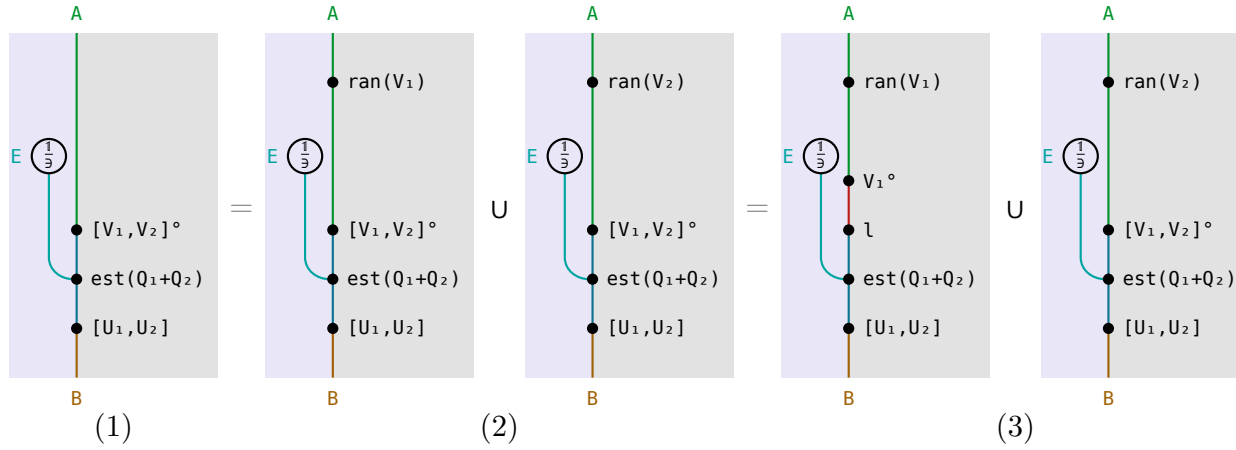
1	$H^\circ \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$	$H^\circ = h^\circ F(H^\circ) T$
2	$= h^\circ F(H^\circ) T \stackrel{T^\circ}{\subseteq} \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$	$T \stackrel{T^\circ}{\subseteq} \epsilon$
3	$\sqsubseteq h^\circ F(H^\circ) \epsilon \text{est}(Q) F\left(\frac{H}{\exists} \text{est}(R)\right) h$	$\epsilon \text{est}(R) \subseteq R^\circ$
4	$\sqsubseteq h^\circ F(H^\circ) Q^\circ F\left(\frac{H}{\exists} \text{est}(R)\right) h$	$\text{ThinCondition}(T, h, R, Q)$ $\Rightarrow h^\circ F(H^\circ) Q^\circ \subseteq R^\circ h^\circ F(H^\circ)$



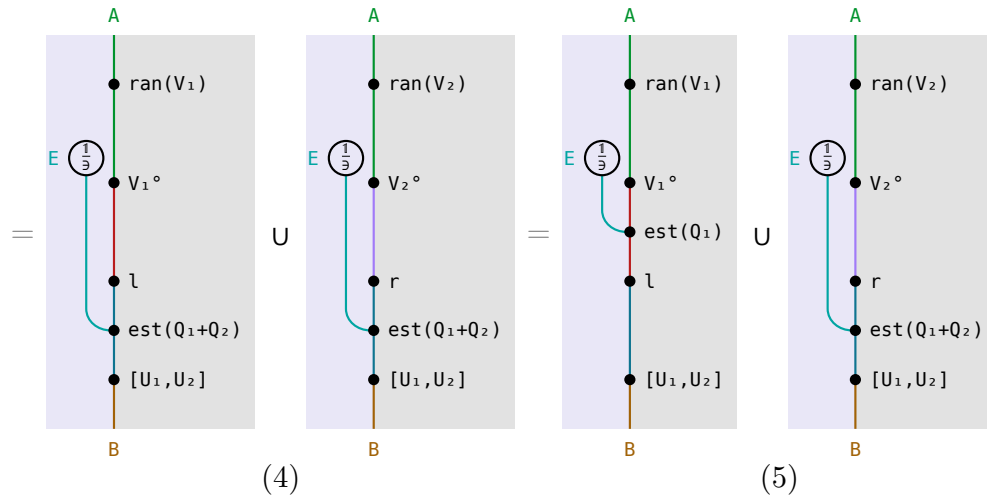
5	$\sqsubseteq R^\circ h^\circ F(H^\circ \frac{H}{\exists} \text{est}(R)) h$	$S^\circ \stackrel{S}{\subseteq} \text{est}(R) \subseteq R^\circ$
6	$\sqsubseteq R^\circ h^\circ F(R^\circ) h$	$\text{Map}(h) \wedge h \text{ Pres } R$ $\Rightarrow h^\circ F(R^\circ) h \subseteq R^\circ$
7	$\sqsubseteq R^\circ R^\circ$	$R R \subseteq R$ $\Rightarrow R^\circ R^\circ \subseteq R^\circ$
8	$\sqsubseteq R^\circ$	

$$V_2 \ V_1^\circ = \emptyset \implies \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} \text{est}(Q_2) U_2) \quad \text{Proposition 10.1}$$

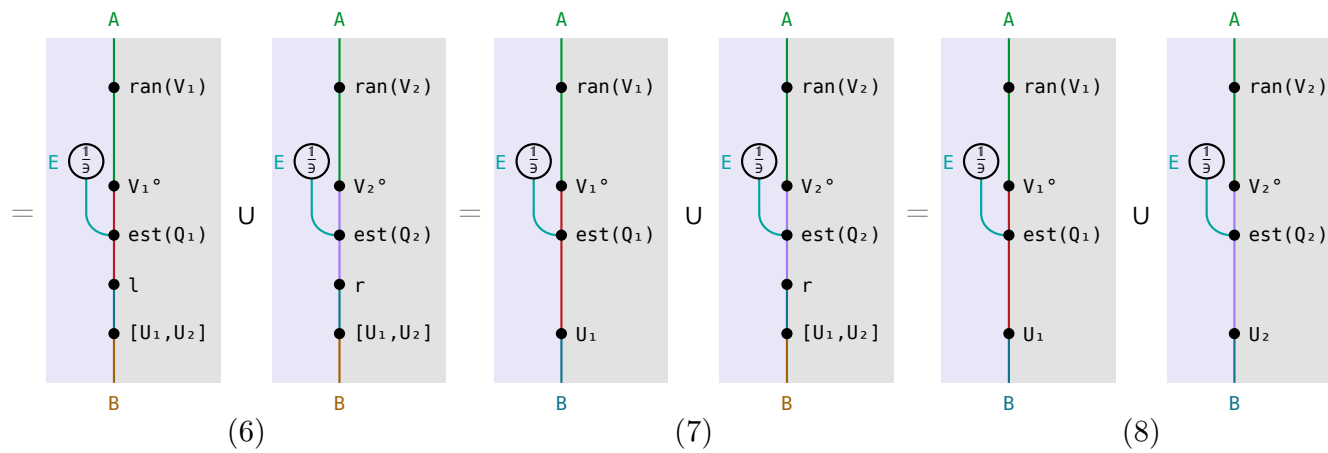
when V_1 and V_2 have disjoint ranges, the greedy step over $[V_1, V_2]$ runs the V_1 step on inputs V_1 reaches and the V_2 step on inputs V_2 reaches



1	$\frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]$	$\frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2] = (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$
2	$= (\text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$	$V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_1) \ \frac{[V_1, V_2]^\circ}{3} = \text{ran}(V_1) \ \frac{V_1^\circ}{3} P(l)$
3	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{3} \text{est}(Q_1 + Q_2) [U_1, U_2])$	$V_2 \ V_1^\circ = \emptyset \implies \text{ran}(V_2) \ \frac{[V_1, V_2]^\circ}{3} = \text{ran}(V_2) \ \frac{V_2^\circ}{3} P(r)$



4	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} P(l) \text{est}(Q_1 + Q_2) [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$	$P(l) \text{est}(Q_1 + Q_2) = \text{est}(Q_1) \ l$
5	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) \ l [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} P(r) \text{est}(Q_1 + Q_2) [U_1, U_2])$	$P(r) \text{est}(Q_1 + Q_2) = \text{est}(Q_2) \ r$



6	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) \ l [U_1, U_2]) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} \text{est}(Q_2) \ r [U_1, U_2])$	$l [R, S] = R$
7	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} \text{est}(Q_2) \ r [U_1, U_2])$	$r [R, S] = S$
8	$= (\text{ran}(V_1) \ \frac{V_1^\circ}{3} \text{est}(Q_1) U_1) \cup (\text{ran}(V_2) \ \frac{V_2^\circ}{3} \text{est}(Q_2) U_2)$	

§10.2. The detab-entab problem

§10.2.1. Detab

(10.2.1a)

	definition	type	meaning
1	$\llbracket [\text{nil}, \text{expand}] \rrbracket = \text{detab}$	$\text{String} \rightarrow \text{String}$	replace every tab by blanks up to the next tab stop; $H = \text{detab}^\circ$
2	$\text{expand}(xs, a) \triangleq \text{if } a = \text{tb} \text{ then } \text{fill}(xs) \text{ else } \text{snoc}(xs, a)$	String	a tab fills the line to the next stop; any other character is appended
3	$\text{fill}(xs) \triangleq xs \# \text{blanks}(n - \text{col}(xs) \% n)$	String	append blanks up to the next multiple of n columns
4	$\text{col}(\text{wrap}(a)) = 0, \text{col}(\text{snoc}(x, a)) = \text{if } a = \text{nl} \text{ then } 0 \text{ else } \text{col}(x) + 1$	$\text{String} \rightarrow \mathbb{N}$	the column the string ends at: characters since the last newline
5	$xs \text{ R } ys \iff \text{length}(xs) \leq \text{length}(ys)$	$[\text{Char}] \rightarrow [\text{Char}]$	x is no longer than y
6	$a \text{ U } b \iff a = \text{tb} \vee a = b$	$\text{Char} \rightarrow \text{Char}$	a is the tab or $a=b$
7	$xs \text{ V } ys \iff \text{prefix}(xs, ys) \wedge \text{fill}(xs) = \text{fill}(ys)$	$[\text{Char}] \rightarrow [\text{Char}]$	a prefix that fills to the same string
8	$Q \triangleq 1 + (V \times U)$	$F[\text{Char}] \rightarrow F[\text{Char}]$	compare two decompositions by V on the strings and U on the characters
9	$\text{unfill}(\text{wrap}(a)) = [], \text{unfill}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc}(x, a)) \% n \neq 0 \text{ then } \text{unfill}(x) \text{ else } \text{snoc}(x, a)$	$\text{String} \rightarrow \text{String}$	the shortest prefix that fills to the same string
10	$\text{tbc}(\text{wrap}(a)) = 0, \text{tbc}(\text{snoc}(x, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc}(x, a)) \% n \neq 0 \text{ then } \text{tbc}(x) + 1 \text{ else } 0$	$\text{String} \rightarrow \mathbb{N}$	the number of trailing blanks
11	$\text{triple} \triangleq (\text{unfill } \text{entab}, (\text{tbc}, \text{col}))$	$[\text{Char}] \rightarrow \text{String} \times (\mathbb{N} \times \mathbb{N})$	what entab keeps: the unfilled output, its trailing blank count and its column

§10.2.2. Derivation

$$0 < n \Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand } u \ (\pi_1 V^\circ)$$

(10.2.2a)

shortening the output of one **expand** step to a V -smaller string either keeps the whole step (**expand**) or drops its character and shortens its input string ($\pi_1 V^\circ$)

(1)	(2)	(3)	(4)
1 $\text{expand } V^\circ$ 2 $= ((\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc})) V^\circ$ 3 $= (\text{istab } \pi_1 \text{ fill } V^\circ) \cup (\text{nottab } \text{snoc } V^\circ)$		$\text{expand} = (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc})$ $(S \cup T) R = (S R) \cup (T R)$ $0 < n \wedge \text{blank} \neq \text{nl}$ $\Rightarrow \text{fill } V^\circ = \text{fill}$	
(5)	(6)	(7)	(8)
4 $= (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } \text{snoc } V^\circ)$ 5 $\sqsubseteq (\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } (\text{snoc } \cup (\pi_1 V^\circ)))$ 6 $\sqsubseteq \text{expand } \cup (\pi_1 V^\circ)$		$0 < n \wedge \text{blank} \neq \text{nl}$ $\Rightarrow \text{snoc } V^\circ \sqsubseteq \text{snoc } \cup (\pi_1 V^\circ)$ $(\text{istab } \pi_1 \text{ fill}) \cup (\text{nottab } (\text{snoc } \cup X)) \sqsubseteq \text{expand} \cup X$	

$$0 < n \Rightarrow \text{detab } V^\circ \sqsubseteq R^\circ \text{ detab}$$

(10.2.2b)

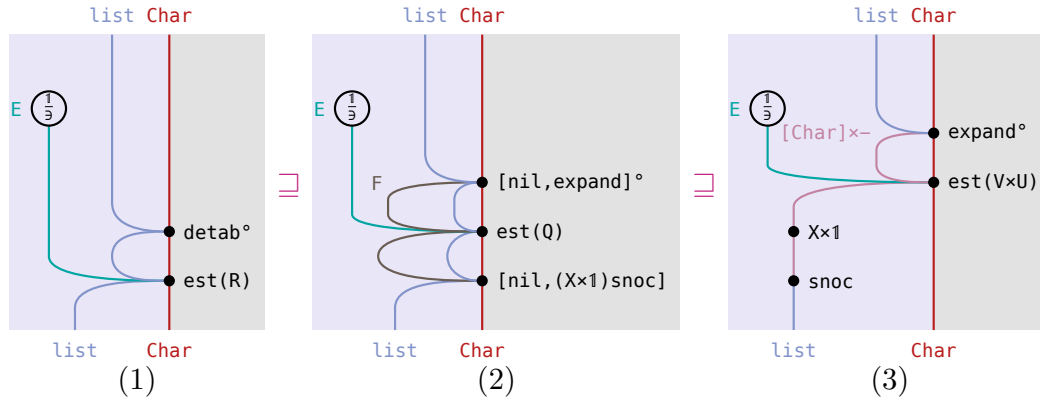
any V -shortening of **detab**'s output is **detab**'s output on an input no longer than the given one

(1)	(2)	(3)	(4)
1 $\text{detab } V^\circ$ 2 $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}] V^\circ$ 3 $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil } V^\circ, \text{expand } V^\circ]$ 4 $= [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } V^\circ]$		$\text{detab} = [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand}]$ $[R, S] Z = [R Z, S Z]$ $\text{nil } V^\circ = \text{nil}$ (10.2.2a) $0 < n$ $\Rightarrow \text{expand } V^\circ \sqsubseteq \text{expand } \cup (\pi_1 V^\circ)$	
(5)	(6)	(7)	(8)
5 $\sqsubseteq [\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } \cup (\pi_1 V^\circ)]$ 6 $= \text{detab } \cup (\text{snoc}^\circ (\text{detab} \times 1) \pi_1 V^\circ)$ 7 $= \text{detab } \cup (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ)$ 8 $\sqsubseteq R^\circ \text{ detab}$		$[\text{nil}, \text{snoc}]^\circ F(\text{detab}) [\text{nil}, \text{expand } \cup X] = \text{detab } \cup (\text{snoc}^\circ (\text{detab} \times 1) X)$ $\text{snoc}^\circ (\text{detab} \times 1) \pi_1 Y = \text{snoc}^\circ \pi_1 \text{ detab } Y$ $0 < n$ $\Rightarrow \text{detab } \cup (\text{snoc}^\circ \pi_1 \text{ detab } V^\circ) \sqsubseteq R^\circ \text{ detab}$	

$$0 < n \Rightarrow (\mu X : \frac{[nil, expand]^{\circ}}{3} \text{ est}(Q) F(X) [nil, snoc]) \sqsubseteq \frac{detab^{\circ}}{3} \text{ est}(R) \quad (10.2.2c)$$

the shortest input **detab** expands to the given output is one pass along that output, holding each blank back and caching the held blanks in for a tab wherever the column reaches a tab stop

the specification — $\text{detab}(\text{wrap}(a)) = []$, $\text{detab}(\text{snoc}(x, a)) = \text{expand}(\text{detab}(x), a)$, $\langle [nil, expand] \rangle = \text{detab}; \text{detab}$ $\text{entab} = 1$ and nothing shorter does



$$1 \quad \frac{detab^{\circ}}{3} \text{ est}(R)$$

$$2 \quad \sqsubseteq \frac{[nil, expand]^{\circ}}{3} \text{ est}(Q) F(X) [nil, snoc]$$

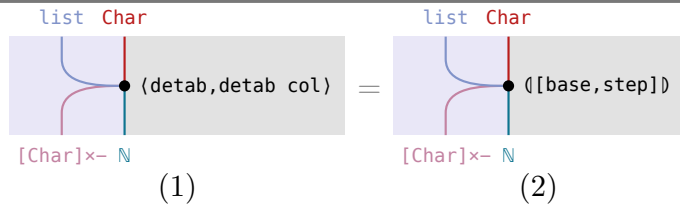
$$3 \quad \sqsubseteq \frac{expand^{\circ}}{3} \text{ est}(V \times U) (X \times 1) \text{ snoc}$$

at $Q \triangleq 1 + (V \times U)$: one character of input is decided at each step. $F(T, R) \alpha \sqsubseteq \alpha R$. $\text{detab prefix} \sqsubseteq R^{\circ} \text{ detab}$ is FALSE, — at $n=8$, $\text{detab } [a, b, c, d, e, TB] = [a, b, c, d, e, BL, BL, BL]$, whose prefix $[a, b, c, d, e, BL, BL]$ is longer than any input giving it, and $\text{detab } V^{\circ} \sqsubseteq R^{\circ} \text{ detab}$ holds. $\text{expand } V^{\circ} \sqsubseteq \text{expand } U (\pi_1 V^{\circ})$ — shortening the output either leaves the last step alone or discards it

Proposition 10.1: **nil** and **expand** have disjoint ranges. The greedy step is to emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop

$$\langle \text{detab}, \text{detab col} \rangle = \langle [base, step] \rangle \quad (10.2.2d)$$

one pass over the input carries the output so far together with its column



$$1 \quad \langle \text{detab}, \text{detab col} \rangle$$

$$2 \quad = \langle [base, step] \rangle$$

both components are folds over the same input; **step** on $(\text{detab } x, \text{col}(\text{detab } x))$ is $(\text{detab}, \text{col detab})$ one character further — not a tabulated row

Exercise 10.1

$$\langle [base, step] \rangle (\text{convert}(xs), r) \Leftrightarrow r = \text{loop}(\text{step})(\langle [], 0 \rangle, xs)$$

the fold of $[base, step]$ over the input converted to a snoc-list is the left loop of **step** from **base**

$$\pi_1(\text{loop}(\text{step})(\langle [], c_0 \rangle, xs)) = \text{loop}'((f, g), (c_0, xs))$$

Exercise 10.1: when **step** only appends $f(c, a)$ to the output and moves the state by g , the output of the loop is $\text{loop}'(f, g)$, which never carries the output

(10.2.2e)

$$a \neq tb \Rightarrow \text{expand}(ys, b) = \text{snoc}(xs, a) \Leftrightarrow \text{fill}(ys) = \text{snoc}(xs, a) \wedge b = tb \vee (ys, b) = (xs, a)$$

expand produces $x \# [a]$ from a tab after a string that fills to it, or from x and a

$$0 < n \Rightarrow (\exists ys. \text{fill}(ys) = \text{snoc}(xs, a)) \Leftrightarrow a = \text{blank} \wedge \text{col}(\text{snoc}(xs, a)) \% n = 0$$

a string is a **fill** exactly when it ends in a blank on a tab stop

(10.1)

$$\text{entab}(xs) = \text{entab}(\text{unfill}(xs)) \# \text{blanks}(\text{tbc}(xs))$$

(10.1): the output of **entab** is that of **unfill** x followed by the trailing blanks held back

$$\text{tbc}[] = 0$$

the empty string has no trailing blanks

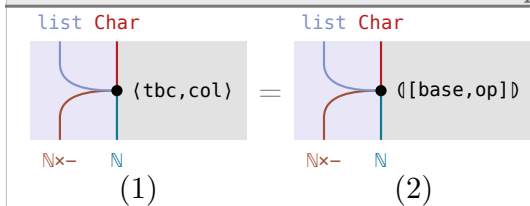
$$\text{tbc}(\text{snoc}(xs, a)) = \text{if } a = \text{blank} \wedge \text{col}(\text{snoc}(xs, a)) \% n \neq 0 \text{ then } \text{tbc}(xs) + 1 \text{ else } 0$$

a blank off a tab stop adds one held blank; anything else releases them all

(tbc,col) = ([base,op])

(10.2.2f)

one pass counts the held blanks and the column together



1 (tbc,col)

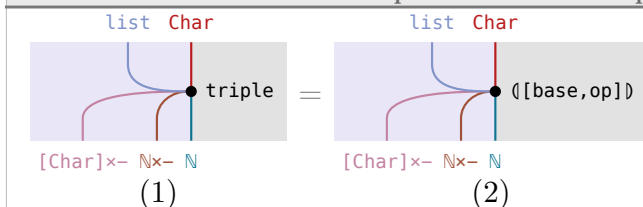
2 = ([base,op])

op on (tbc x,col x) is (tbc,col) one character further,
 from the equations for tbc — (10.1) entab(xs) =
 entab(unfill(xs))#blanks(tbc(xs)), tbc[] = 0,
 tbc(snoc(xs,a)) = if a = blank $\wedge$
 col(snoc(xs,a)) % n $\neq$ 0 then tbc(xs) + 1 else 0

triple = ([base,op])

(10.2.2g)

one pass carries the output up to the held blanks, their count and the column



1 triple

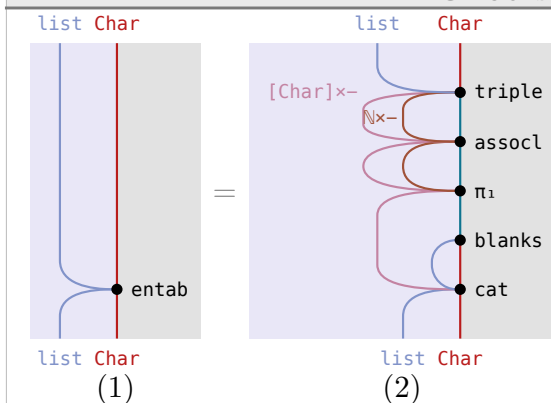
2 = ([base,op])

op on triple x is triple one character further: (10.1) where
 the blanks are released — (10.1) entab(xs) =
 entab(unfill(xs))#blanks(tbc(xs)), tbc[] = 0,
 tbc(snoc(xs,a)) = if a = blank $\wedge$
 col(snoc(xs,a)) % n $\neq$ 0 then tbc(xs) + 1 else 0

entab = triple assocl π_1 (1 $\times$ blanks) cat

(10.2.2h)

entab is triple with the held blanks appended to its output



1 entab

2 = triple assocl π_1 (1 $\times$ blanks) cat

(10.1) — (10.1) entab(xs) = entab(unfill(xs))
 #blanks(tbc(xs)), tbc[] = 0, tbc(snoc(xs,a)) =
 if a = blank $\wedge$ col(snoc(xs,a)) % n $\neq$ 0 then
 tbc(xs) + 1 else 0

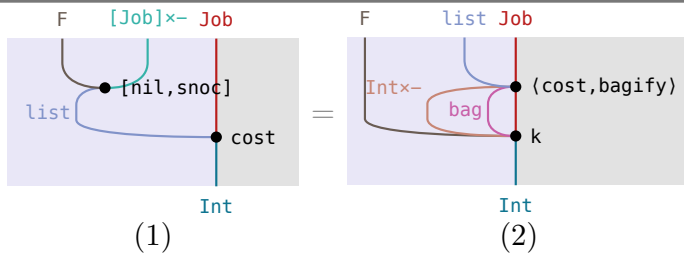
§10.3. The minimum tardiness problem

(10.3a)

	definition	type	meaning
1	$\text{bag}(\text{Job}) \triangleq \text{Quotient}(\text{perm})$	$\mathcal{A}$	a bag of jobs: a list up to reordering
2	$\text{snag}(p) \triangleq \pi_2(p) :: \pi_1(p)$	$\text{bag}(\text{Job})$	put one job into a bag
3	$\langle [\text{nil}, \text{snag}] \rangle = \text{bagify}$	$[\text{Job}] \rightarrow \text{bag}(\text{Job})$	forget the order of a schedule; $H = \text{bagify}^\circ$ lists every schedule of a bag
4	$\text{penalty}(xs, j) \triangleq (\text{sum}(\text{list}(\text{ct}))(xs) + \text{ct}(j) - \text{dt}(j)) \text{wt}(j)$	Int	how late job j finishes after the schedule xs , times its weight
5	$\text{cost}(\text{wrap}(a)) = 0, \text{cost}(\text{snoc}(x, a)) = \text{bmax}(\text{cost}(x), \text{penalty}(x, a))$	$[\text{Job}] \rightarrow \text{Int}$	the largest penalty of any job in the schedule
6	$\text{bmax}(a, b) \triangleq \text{if } a \leq b \text{ then } b \text{ else } a$	Int	the larger of two numbers
7	$R \triangleq \text{cost} \leq \text{cost}^\circ$	$[\text{Job}] \rightarrow [\text{Job}]$	xs costs no more than ys
8	$p \text{ add } ys \iff \text{add}(\pi_2(p), ys)$	$[\text{Job}] \times \text{Job} \rightarrow [\text{Job}]$	insert j anywhere in xs ; $\text{perm} \triangleq \text{bagify}$ $\text{bagify}^\circ = \langle [\text{nil}, \text{add}] \rangle$
9	$k \triangleq [\text{nil}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{ penalty})) \text{ bmax}]$	$F(\text{Int} \times \text{bag}(\text{Job})) \rightarrow \text{Int}$	the cost of a schedule from the cost of its front and the penalty of its last job
10	$f(l(\text{val})) = 0, f(r(p)) = (\text{bagify}^\circ \times 1) \text{ penalty}(p)$	$F(\text{bag}(\text{Job})) \rightarrow \text{Int}$	the penalty of the last job, read off the bag before it
11	$u Q v \iff f(u) \leq f(v)$	$F(\text{bag}(\text{Job})) \rightarrow F(\text{bag}(\text{Job}))$	the last job of u gets no larger penalty than that of v
12	$p Q' q \iff (\text{bagify}^\circ \times 1) \text{ penalty}(p) \leq (\text{bagify}^\circ \times 1) \text{ penalty}(q)$	$\text{bag}(\text{Job}) \times \text{Job} \rightarrow \text{bag}(\text{Job}) \times \text{Job}$	Q on the snag summand: compare the penalties of the two last jobs
13	$g \triangleq [\text{zero}, \text{penalty}]$	$F[\text{Job}] \rightarrow \text{Int}$	(10.5): the penalty of the last job, zero for the empty schedule
14	$m \triangleq [\text{zero}, \pi_1 \text{ cost}]$	$F[\text{Job}] \rightarrow \text{Int}$	(10.6), B&dM's h renamed as in $(\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j))$ $\implies (\mu X : \frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) F(X) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{bagify}^\circ}{\exists}$ $\text{est}(R)$: the cost of the schedule before the last job

$$[\text{nil}, \text{snoc}] \text{ cost} = F(\langle \text{cost}, \text{bagify} \rangle) k \quad (10.3b)$$

the cost of a schedule is got from the cost and the bag of the schedule before its last job, and that job



1 $[\text{nil}, \text{snoc}] \text{ cost}$

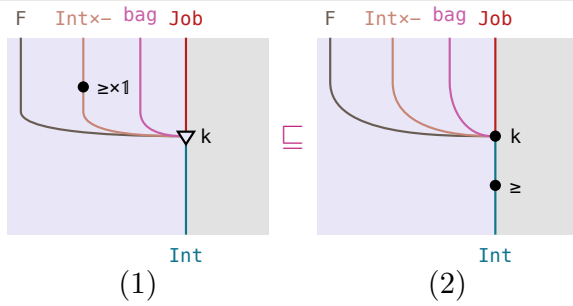
$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$, and $\text{penalty}(xs, j)$ reads only the bag of xs — $\text{penalty}(xs, j) \triangleq (\text{sum}(\text{list}(\text{ct}))(xs) + \text{ct}(j) - \text{dt}(j)) \text{wt}(j)$

2 $= F(\langle \text{cost}, \text{bagify} \rangle) k$

$$F(\geq \times 1) k \sqsubseteq k \geq$$

Exercise 10.5

a larger cost before the last job gives a larger cost after it



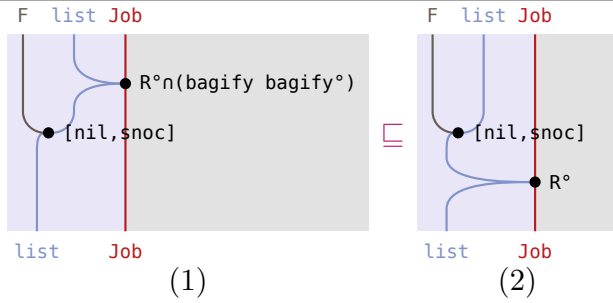
1 $F(\geq \times 1) k$

bmax is monotone in its first argument

2 $\sqsubseteq k \geq$

$$F(R^\circ \text{ n } (\text{bagify } \text{bagify}^\circ)) [\text{nil}, \text{snoc}] \sqsubseteq [\text{nil}, \text{snoc}] R^\circ \quad (10.2)$$

improving the schedule before the last job, among schedules of the same bag, improves the whole schedule



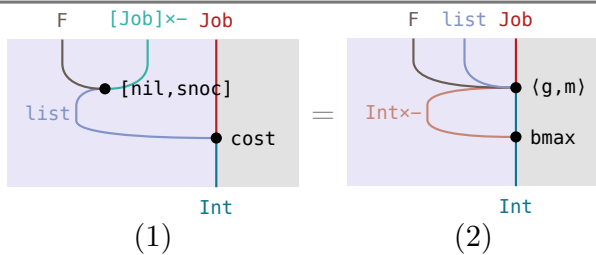
1 $F(R^\circ \text{ n } (\text{bagify } \text{bagify}^\circ)) [\text{nil}, \text{snoc}]$

Proposition 9.3 at $S \triangleq \text{bagify}$, $\leq \triangleq$ — Proposition 9.3
 $\text{Map}(\text{cost}) \wedge \text{Simple}(S) \wedge h \text{ cost} = F(\langle \text{cost}, S \rangle) k$
 $\wedge F(\leq \times 1) k \sqsubseteq k \leq$
 $\Rightarrow F((\text{cost} \leq \text{cost}^\circ) \text{ n } (S \leq S^\circ)) h \sqsubseteq h \text{ cost} \leq \text{cost}^\circ$, with $[\text{nil}, \text{snoc}] \text{ cost} = F(\langle \text{cost}, \text{bagify} \rangle) k$
 and Exercise 10.5 $F(\geq \times 1) k \sqsubseteq k \geq$

2 $\sqsubseteq [\text{nil}, \text{snoc}] R^\circ$

$$[\text{nil}, \text{snoc}] \text{ cost} = \langle g, m \rangle \text{ bmax} \quad (10.4)$$

the cost of a schedule is the larger of the penalty of its last job and the cost of the schedule before it



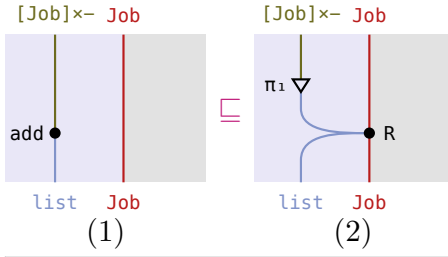
1 $[\text{nil}, \text{snoc}] \text{ cost}$

$\text{cost}(xs \# [j]) = \text{bmax}(\text{cost } xs, \text{penalty}(xs, j))$ and bmax commutes — $\text{cost}(\text{wrap}(a)) = 0$,
 $\text{cost}(\text{snoc}(x, a)) = \text{bmax}(\text{cost}(x), \text{penalty}(x, a))$

2 $= \langle g, m \rangle \text{ bmax}$

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow add \sqsubseteq \pi_1 R \quad (10.7)$$

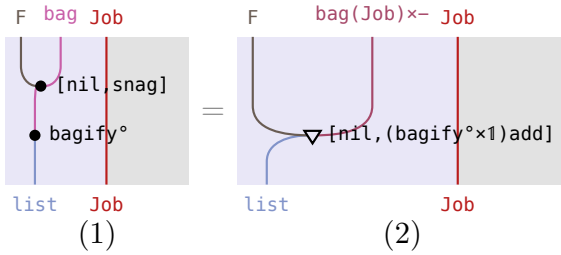
putting one more job anywhere into a schedule never lowers its cost



1	add	induction on where j goes: every job after it starts later, so no penalty falls
2	$\sqsubseteq \pi_1 R$	

$$[nil, snag] \text{ bagify}^\circ = [nil, (\text{bagify}^\circ \times 1) \text{ add}] \quad (10.8)$$

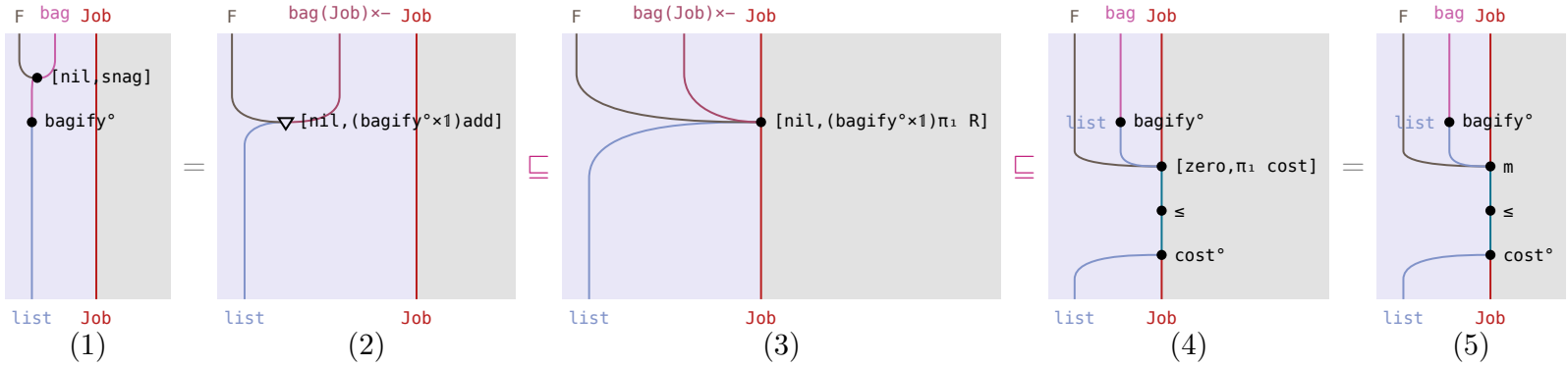
the orderings of a bag with one more job are the orderings of the bag with that job put in anywhere



1	$[nil, snag] \text{ bagify}^\circ$	take out the last copy of j and put it back — $p \text{ add } ys \iff \text{add}(\pi_2(p), ys)$
2	$= [nil, (\text{bagify}^\circ \times 1) \text{ add}]$	

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow [nil, snag] \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) m \leq \text{cost}^\circ \quad (10.3c)$$

an ordering of a bag with one more job costs at least the cost of the ordering of the bag before its last job



1	$[nil, snag] \text{ bagify}^\circ$	(10.8) $[nil, snag] \text{ bagify}^\circ = [nil, (\text{bagify}^\circ \times 1) \text{ add}]$
2	$= [nil, (\text{bagify}^\circ \times 1) \text{ add}]$	(10.7) $(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow add \sqsubseteq \pi_1 R$
3	$\sqsubseteq [nil, (\text{bagify}^\circ \times 1) \pi_1 R]$	$[nil, \pi_1 R] \sqsubseteq [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$
4	$\sqsubseteq F(\text{bagify}^\circ) [zero, \pi_1 \text{ cost}] \leq \text{cost}^\circ$	
5	$= F(\text{bagify}^\circ) m \leq \text{cost}^\circ$	

$$(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j)) \Rightarrow (Q \cap ([nil, snag] [nil, snag]^\circ)) F(bagify^\circ) [nil, snoc] \sqsubseteq F(bagify^\circ) [nil, snoc] R \quad (10.3)$$

a schedule of a bag ending in a job of least penalty is no worse than any schedule of the same bag

(1)	(2)	(3)	(4)
1 $(Q \cap ([nil, snag] [nil, snag]^\circ)) F(bagify^\circ) [nil, snoc]$	$(S \cap T) R \sqsubseteq (S R) \cap (T R)$		
2 $\sqsubseteq Q F(bagify^\circ) [nil, snoc] \cap ([nil, snag] [nil, snag]^\circ F(bagify^\circ) [nil, snoc])$	$[nil, snag]^\circ F(bagify^\circ) [nil, snoc] = bagify^\circ$		
3 $= (Q F(bagify^\circ) [nil, snoc]) \cap ([nil, snag] bagify^\circ)$	(10.3c) $(\forall j. 0 \leq ct(j)) \wedge (\forall j. 0 \leq wt(j))$		
	$\Rightarrow [nil, snag] bagify^\circ \sqsubseteq F(bagify^\circ) m \leq cost^\circ$		
4 $\sqsubseteq (Q F(bagify^\circ) [nil, snoc]) \cap (F(bagify^\circ) m \leq cost^\circ)$	$(Q F(bagify^\circ) [nil, snoc]) \cap (F(bagify^\circ) M) \sqsubseteq$		
	$F(bagify^\circ) ((F(bagify^\circ) Q F(bagify^\circ)) \cap (M [nil, snoc]^\circ)) [nil, snoc]$		
(5)	(6)	(7)	(8)
5 $\sqsubseteq F(bagify^\circ) ((F(bagify^\circ) Q F(bagify^\circ)) \cap (m \leq cost^\circ [nil, snoc]^\circ)) [nil, snoc]$	$F(bagify^\circ) Q F(bagify^\circ) = g \leq g^\circ$		
6 $= F(bagify^\circ) ((g \leq g^\circ) \cap (m \leq cost^\circ [nil, snoc]^\circ)) [nil, snoc]$	$(R S)^\circ = S^\circ R^\circ$		
7 $= F(bagify^\circ) ((g \leq g^\circ) \cap (m \leq ([nil, snoc] cost)^\circ)) [nil, snoc]$	$\langle X, Y \rangle \langle R, S \rangle^\circ = \langle X R^\circ \rangle \cap \langle Y S^\circ \rangle$		
	(10.3d) $\langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ$		
8 $= F(bagify^\circ) \langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ [nil, snoc]$	$[nil, snoc] \sqsubseteq [nil, snoc] R$		
9 $\sqsubseteq F(bagify^\circ) [nil, snoc] R$			

$$\langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ [nil, snoc] \sqsubseteq [nil, snoc] R \quad (10.3d)$$

a job whose penalty is at most the last penalty, put after a schedule costing at most the cost before the last job, gives a schedule costing at most the whole

(1)

(2)

(3)

1 $\langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ [nil, snoc]$	$1 \sqsubseteq cost cost^\circ$
2 $\sqsubseteq \langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ [nil, snoc] cost cost^\circ$	$[nil, snoc] cost = \langle g, [nil, snoc] cost \rangle bmax$
3 $= \langle g \leq, m \leq \rangle \langle g, [nil, snoc] cost \rangle^\circ \langle g, [nil, snoc] cost \rangle bmax cost^\circ$	$\langle g, [nil, snoc] cost \rangle^\circ \langle g, [nil, snoc] cost \rangle \sqsubseteq 1$

(4)

(5)

(6)

(7)

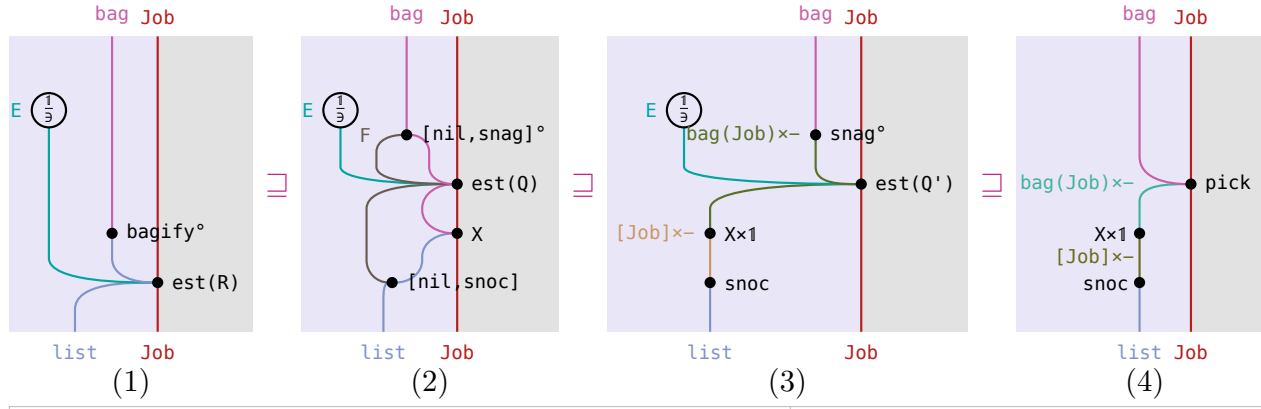
4 $\sqsubseteq \langle g \leq, m \leq \rangle bmax cost^\circ$	$\langle g \leq, m \leq \rangle bmax \sqsubseteq \langle g, m \rangle bmax \leq$
5 $\sqsubseteq \langle g, m \rangle bmax \leq cost^\circ$	(10.4) $[nil, snoc] cost = \langle g, m \rangle bmax$
6 $= [nil, snoc] cost \leq cost^\circ$	$R = cost \leq cost^\circ$
7 $= [nil, snoc] R$	

$$\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick} (\text{schedule} \times 1) \text{ snoc}), (\forall j. 0 \leq \text{ct}(j)) \wedge (\forall j. 0 \leq \text{wt}(j)) \Rightarrow (\mu X : \quad (10.3e)$$

$$\frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) \text{ F}(X) [\text{nil}, \text{snoc}] \sqsubseteq \frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$$

an ordering of the given bag with least maximum penalty is got by taking a job of least penalty out of the bag, putting it last, and scheduling what is left the same way

the specification — $\langle [\text{nil}, \text{snag}] \rangle = \text{bagify}$



1	$\frac{\text{bagify}^\circ}{\exists} \text{ est}(R)$	No greedy reduce exists — one would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole
2	$\sqsubseteq \frac{[\text{nil}, \text{snag}]^\circ}{\exists} \text{ est}(Q) \text{ F}(X) [\text{nil}, \text{snoc}]$	Proposition 10.1: nil and snag have disjoint ranges
3	$\sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q') (X \times 1) \text{ snoc}$	$\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\exists} \text{ est}(Q')$, a partial function, quadratic in the number of jobs
4	$\sqsubseteq \text{pick} (X \times 1) \text{ snoc}$	

§10.4. The TeX problem

(10.4a)

	definition	type	meaning
1	$\text{intern} \triangleq \text{val round}$	$[\text{Digit}] \rightarrow [0, 2^{16})$	the value of a decimal fraction, rounded to 16 binary places
2	$\text{val} \triangleq ([\text{zero}, \text{shift}])$	$[\text{Digit}] \rightarrow \text{Real}$	the real number a digit list denotes after the decimal point
3	$\text{shift}(d, r) \triangleq (d + r)/10$	$\text{Digit} \times \text{Real} \rightarrow \text{Real}$	put digit d in front of r : $(d+r)/10$
4	$\text{round}^\circ = \text{interval inrange}$	$\text{Real} \rightarrow [0, 2^{16})$	the nearest integer to $2^{16}r$: $\text{round } r = n$ iff $2n-1 < 2^{17}r < 2n+1$
5	$\text{interval}(n) \triangleq ((2n-1)/w, (2n+1)/w)$	$[0, 2^{16}) \rightarrow \text{Interval}$	the open interval $((2n-1)/2^{17}, (2n+1)/2^{17})$ of the reals that round to n
6	$p \text{ inrange } r \iff \text{lo}(p) < r \wedge r < \text{hi}(p)$	$\text{Interval} \rightarrow \text{Real}$	r lies strictly inside (a, b)
7	$\text{Legal}(p) \triangleq 0 < \text{hi}(p) \wedge \text{hi}(p) < 1 \wedge \text{lo}(p) < \text{hi}(p)$	Prop	(10.9): $0 < b < 1$ and $a < b$; Interval is the legal pairs
8	$x \text{ arb } p \iff \text{lo}(p) < 0 \wedge 0 < \text{hi}(p)$	$1 \rightarrow \text{Interval}$	any legal interval
9	$\text{step}(p) \triangleq (\text{shift}(\pi_1(p), \text{lo}(\pi_2(p))), \text{shift}(\pi_1(p), \text{hi}(\pi_2(p))))$	$\text{Digit} \times \text{Interval} \rightarrow \text{Interval}$	put digit d in front of both ends: $((d+a)/10, (d+b)/10)$
10	$H \triangleq ([\text{arb}, \text{step}])^\circ$	$\text{Interval} \rightarrow [\text{Digit}]$	every digit list whose interval the given one is
11	$xs \text{ R } ys \iff \text{length}(xs) \leq \text{length}(ys)$	$[\text{Digit}] \rightarrow [\text{Digit}]$	x is no longer than y
12	$l \triangleq \text{inl}$	$1 \rightarrow \text{FX}$	the injection of the empty case into $\text{FX} = 1 + (\text{Digit} \times X)$
13	$r \triangleq \text{inr}$	$\text{Digit} \times X \rightarrow \text{FX}$	the injection of the digit case into $\text{FX} = 1 + (\text{Digit} \times X)$
14	$! \triangleq \rightarrow ()$	$\text{Digit} \times \text{Interval} \rightarrow 1$	the one map to 1
15	$Q \triangleq (l^\circ !^\circ r) \cup 1$	$F(\text{Interval}) \rightarrow F(\text{Interval})$	u is the empty case and z a digit case, or $u=z$
16	$w \triangleq 131072$	Int	2^{17}

 $([\text{arb}, \text{step}]) \sqsubseteq \text{val inrange}^\circ$

(10.4b)

every interval got by folding a decimal's digits with $[\text{arb}, \text{step}]$ has the decimal's value strictly inside it

(1)	(2)	(3)	(4)	(5)
1	$F(\text{inrange}^\circ) [\text{arb}, \text{step}]$	$F(S) [T, U] = [T, (1 \times S) U]$		
2	$= [\text{arb}, (1 \times \text{inrange}^\circ) \text{step}]$	$(1 \times \text{inrange}^\circ) \text{step} \sqsubseteq \text{shift inrange}^\circ$		
3	$\sqsubseteq [\text{arb}, \text{shift inrange}^\circ]$	$\text{zero inrange}^\circ = \text{arb}$		
4	$= [\text{zero inrange}^\circ, \text{shift inrange}^\circ]$	$[R, S] Z = [R Z, S Z]$		
5	$= [\text{zero}, \text{shift}] \text{inrange}^\circ$			

$$Q \ F(X) \ \alpha \sqsubseteq F(X) \ \alpha \ R$$

(10.4c)

choosing by Q among the one-step decompositions before building with $F(X)$ and α yields only decimals that are R -related to one built without choosing

list Fin(10) (1)	list Fin(10) (2)	list Fin(10) (3)	list Fin(10) (4)
list Fin(10) (5)		list Fin(10) (6)	
list Fin(10) (7)		list Fin(10) (8)	
list Fin(10) (9)		list Fin(10) (10)	

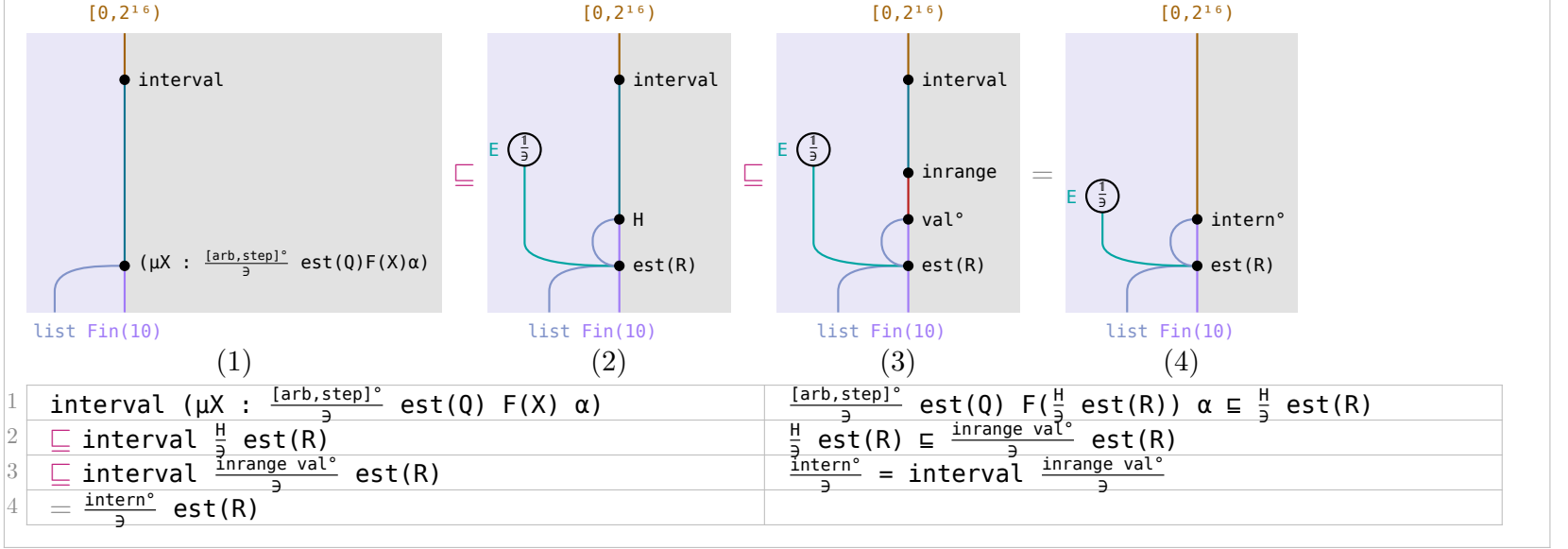
1 $Q \ F(X) \ \alpha$	$Q \triangleq (l^o \ !^o \ r) \cup 1$
2 $\triangleq ((l^o \ !^o \ r) \cup 1) \ F(X) \ \alpha$	$(S \cup T) \ R = (S \ R) \cup (T \ R)$
3 $= (l^o \ !^o \ r \ F(X) \ \alpha) \cup (F(X) \ \alpha)$	$1 \sqsubseteq R$

4 $\sqsubseteq (l^o \ !^o \ r \ F(X) \ \alpha) \cup (F(X) \ \alpha \ R)$	$r \ F(X) \ \alpha \sqsubseteq ! \ l \ \alpha \ R$
5 $\sqsubseteq (l^o \ !^o \ ! \ l \ \alpha \ R) \cup (F(X) \ \alpha \ R)$	$!^o \ ! \sqsubseteq 1$
6 $\sqsubseteq (l^o \ l \ \alpha \ R) \cup (F(X) \ \alpha \ R)$	$l \ F(X) = l$

7 $= (l^o \ l \ F(X) \ \alpha \ R) \cup (F(X) \ \alpha \ R)$	$l^o \ l \sqsubseteq 1$
8 $\sqsubseteq (F(X) \ \alpha \ R) \cup (F(X) \ \alpha \ R)$	$R \cup R = R$
9 $= F(X) \ \alpha \ R$	

$$\text{extern}(n)=f(2n-1,2n+1), \quad \text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{ est}(R) \quad (10.4d)$$

a shortest decimal whose internal representation is the given multiple of 2^{-16} is got by emitting the one digit the interval of admissible reals allows, until that interval contains zero and the empty decimal will do



- $f(a,b) = xs \iff a < 0 \wedge xs = [] \vee \neg a < 0 \wedge (\exists d. \exists ys. d = [10b] \wedge f(10a-d, 10b-d) = ys \wedge xs = [d] \# ys)$
 f gives (a,b) the empty decimal when $a < 0$, and otherwise the digit $d = [10b]$ followed by what it gives $(10a-d, 10b-d)$ (10.4e)
- $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) = f$
the least solution of the greedy recursion of $\text{interval } (\mu X : \frac{[\text{arb}, \text{step}]^\circ}{3} \text{ est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{3} \text{ est}(R)$ is f
- $d = [10b] \wedge e = [10b] \implies d = e$
Exercise 10.12: two digits d and e with $0 < 10b-d < 1$ and $0 < 10b-e < 1$ are equal
- $(0 < q-p \wedge q < w) \wedge p < 0 \implies f(p,q) = []$
the program's first clause: $f(p,q) = []$ when $p < 0$
- $(0 < q-p \wedge q < w) \wedge \neg p < 0 \implies f(p,q) = [10q/w] \# f(10p-w \cdot (10q/w), 10q-w \cdot (10q/w))$
the program's second clause: with $d = (10q) \text{ div } w$, $f(p,q)$ is d followed by $f(10p-w \cdot d, 10q-w \cdot d)$
- $(0 < q-p \wedge q < w) \wedge \text{Reach}(p,q) \implies f(p/w, q/w) = f(p,q)$
on the pairs the program reaches from $\text{interval } n$, the integer f at (p,q) is a value of the rational f at $(p/w, q/w)$
- $\text{interval } f = \text{extern}$
 f after interval is the program extern , whose value at n is $f(2n-1, 2n+1)$

§10. Appendix

§10.1. $P(S) \text{ est}(R) = (\exists S) \cap (\epsilon \setminus (SR^\circ))$

(10.1a)

(p, q) tabulates $W \triangleq (\exists S) \cap (\epsilon \setminus (SR^\circ))$, and $y \triangleq \frac{(p \exists S) \cap (qR)}{\exists}$, a map.

(10.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq y y^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(10.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f^\circ, \cdot \exists \neg \exists} \boxed{\begin{array}{c} q \\ (p \exists S) \cap (qR) \end{array}} \xleftarrow{\Delta \neg \cap, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

(10.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \exists, ^\circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple} \\ \\ \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg \cap, \cdot T \neg / T, ^\circ, f^\circ \cdot \neg f^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array} \\ \\ \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \exists, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple} \\ \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) \cap (qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq q q^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{^\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half} \end{array}$$

§10.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{dom}(\text{est}(R))) \text{ union est}(R)$

(10.2a)

$$\boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)}$$

$$\begin{array}{ccc} \xrightarrow{\text{P a relator}} \boxed{P(\text{dom}(\text{est}(R)))P(\text{est}(R)) \text{ est}(R)} & \xrightarrow{(7.11) \ R \ R \sqsubseteq R} & \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ \Rightarrow P(\text{est}(R)) \text{ est}(R) \sqsubseteq \text{union est}(R) & & \end{array}$$

(10.2b)

$$\begin{array}{ccc} \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} & \xleftarrow{\text{P}(\text{dom}(\text{est}(R))) \text{ union est}(R) \sqsubseteq \text{est}(R)} & \boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ \boxed{P(\text{est}(R)) \text{ est}(R)} & \xRightarrow{\text{est}(R)} & \boxed{\text{est}(R)} \\ \Rightarrow P(S) \text{ est}(R) = (\exists S) \cap (\epsilon \setminus (S R^\circ)) \text{ at } S := \text{est}(R), & & \boxed{\epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \\ \Delta \vdash n, T \vdash \neg T \setminus & & \boxed{\text{est}(R) R^\circ} \end{array}$$

$$\boxed{P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \xleftarrow{\text{est}(R)} \boxed{\text{union est}(R)}$$

$$\begin{array}{ccc} \xleftarrow{T(\text{UnV}) \sqsubseteq T \text{UnTV}, \cdot \exists \vdash \frac{}{\exists}, (7.3) \text{ union } (\epsilon \setminus R) = \epsilon \setminus (\epsilon \setminus R)} & \boxed{(\exists \exists) n (\epsilon \setminus (\epsilon \setminus R^\circ))} & \xleftarrow{\text{modular law}} \boxed{\exists (\exists n (\epsilon \setminus (\epsilon \setminus R^\circ)))} \\ & \boxed{\text{est}(R)} & \boxed{\exists (\exists n (\epsilon \setminus R^\circ))} \\ & & \text{count of } T \cdot \neg T \setminus \end{array}$$

$$\boxed{\epsilon P(\text{dom}(\text{est}(R))) \text{ union est}(R)} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\text{dom}(\text{est}(R)) \epsilon \text{ union est}(R)}$$

$$\begin{array}{ccc} \xleftarrow{\text{dom}(\text{est}(R)) \sqsubseteq \text{est}(R) \text{ est}(R)^\circ} & \boxed{\text{est}(R) \text{ est}(R)^\circ \epsilon \text{ union est}(R)} & \xleftarrow{\text{est}(R)^\circ \sqsubseteq \epsilon, \epsilon \epsilon \text{ union} \sqsubseteq \epsilon} \boxed{\text{est}(R) \epsilon \text{ est}(R)} \\ & \boxed{\text{est}(R) R^\circ} & \boxed{\text{est}(R) R^\circ} \\ & & \text{UP of est} \end{array}$$

§10.3. Shortest paths on a cylinder, on lists

(10.3a)

	definition	type	note
1	$F(A,X)=A+A \times X$	$\mathcal{A} \rightarrow \mathcal{A}$	The base functor: one square of the new column, alone or in front of the path so far.
2	$L = \text{list}^+$	$\mathcal{A} \rightarrow \mathcal{A}$	A path is a non-empty list of squares, one per column crossed.
3	α the initial algebra	$F(\mu F) \rightarrow \mu F$	$\alpha(5)=[5]$ and $\alpha(1,[5])=[1,5]$: a path is started by one square, or extended by one.
4	N the n-tuple relator	$\mathcal{A} \rightarrow \mathcal{A}$	One component per row of a column, so the fold carries n answers at once.
5	$R \triangleq \text{sum} \leq \text{sum}^\circ$	$[\mathbb{N}] \rightarrow [\mathbb{N}]$	The cost of a path, which the cheapest minimises.
6	setify	$A[n] \rightarrow PA$	$\text{setify}(1,2,3,4)=\{1,2,3,4\}$ — which row a component came from is forgotten.
7	moves	$A[n] \rightarrow P(A[n])$	$\text{moves}(x)=\{\text{up}(x), x, \text{down}(x)\}$ — rotated up, unrotated, rotated down.
8	trans	$P(A[n]) \rightarrow (P\ A)[n]$	$\text{trans}\{(a,b,c), (x,y,z)\} = (\{a,x\}, \{b,y\}, \{c,z\})$ — component k of the result is the set of the k-th components.
9	zip	$F(a[n]) \rightarrow (F\ a)[n]$	$\text{zip}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = ((1, \{[5]\}), (2, \{[6]\}), (3, \{[7]\}), (4, \{[8]\}))$.
10	$\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$	$F(PA) \rightarrow PFA$	$\text{cp}(1, \{[5], [6], [8]\}) = \{(1, [5]), (1, [6]), (1, [8])\}$.
11	$\text{gen} \triangleq F(1, \text{moves trans } N(\text{union}))$ $\text{zip } N(\text{cp}(F(A, -))\ E(\alpha))$	$F(NA, N(P(TA))) \rightarrow N(P(TA))$	$\text{gen}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}))$ is worked out in $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -))\ E(\alpha))$.
12	$\text{paths} \triangleq (\text{gen})\ \text{setify}\ \text{union}$	$T(NA) \rightarrow P(TA)$	$\text{paths}[(1,2,3,4), (5,6,7,8)]$ is the union of $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -))\ E(\alpha))$'s four sets: 12 paths, 3 from each entry row.
13	the specification $\text{paths est}(R)$	$[\mathbb{N}[n]] \rightarrow [\mathbb{N}]$	A cheapest path from the entry side to the exit side.

§10.3.1. $N(E(L\ A))$

(10.3.1a)

```

1  5
2  6      [(1,2,3,4), (5,6,7,8)] : L N Nat
3  7
4  8

```

(10.3.1b)

```
gen  $\triangleq$  F(1, moves trans N(union)) zip N(cp(F(A, -)) E( $\alpha$ ))
```

```
(gen)[(5,6,7,8)] = ({[5]}, {[6]}, {[7]}, {[8]})
```

```
gen((1,2,3,4), ({[5]}, {[6]}, {[7]}, {[8]})) =
( {[1,5]}, {[1,6]}, {[1,8]} ,
  {[2,5]}, {[2,6]}, {[2,7]} ,
  {[3,6]}, {[3,7]}, {[3,8]} ,
  {[4,5]}, {[4,7]}, {[4,8]} )
```

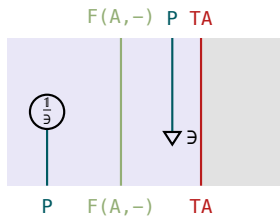
(10.3.1c)

1	N	the 4-tuple, one component per row of the entry column
2	E	the set of paths that can start in that row
3	L	a path is the non-empty list of the squares it crosses

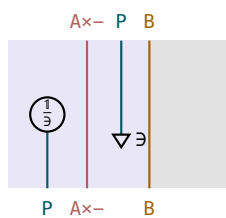
row 1's set has no $[1,7]$ — from row 1 only rows 4, 1 and 2 are reachable — and it does have $[1,8]$, because the cylinder glues the bottom row to the top. That is $\text{moves trans } N(\text{union})$ of $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$: component k collects the paths of rows $k-1, k, k+1$.

§10.3.2. The cross product $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$

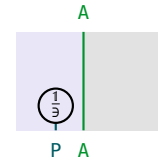
(10.3.2a)



cp on all of $F(A, E(L A))$



the $A \times -$ summand: $\exists$ picks one path, $1 \times \exists$ collects the A summand, no E to distribute: $1 \times \exists$ alone, $a \mapsto \{a\}$



(10.3.2b)

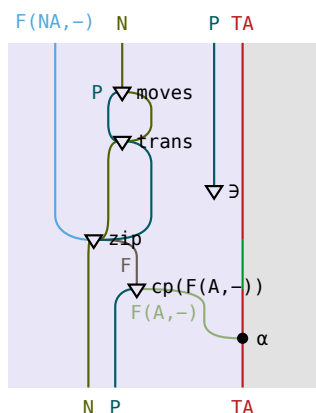
type	note
1 $F(A, -) = A + A \times -$	one wire carries the whole functor, both summands with it; the $\exists$ under it is $F(1, \exists)$
2 $\exists : E(L A) \rightarrow L A$	the $\exists$ inside $F(1, \exists)$: a set of paths, one of them — $L A$ is one path, A one square
3 $F(1, \exists) = 1 + 1 \times \exists$	$: A + A \times E(L A) \rightarrow A + A \times L A$, the relator acting on each summand
4 $\exists : E(A + A \times L A) \rightarrow A + A \times L A$	the $\exists$ under the bar: a set of cells, one of them — a different $\exists$
5 $\text{cp} : A + A \times E(L A) \rightarrow E(A + A \times L A)$	$\frac{R}{\exists}$ turns $R : X \rightarrow Y$ into $X \rightarrow E Y$
6 $\text{cp} = \frac{1}{\exists} E(F(1, \exists))$	what the picture draws: the unit makes the outer E , leaving one $\exists$ — $\frac{R}{\exists} = \frac{1}{\exists} E(R)$

(10.3.2c)

$w = (1, \{[5], [6], [8]\}) : A \times E(L A)$ the right summand
 $1 \times \exists : A \times E(L A) \rightarrow A \times L A$ 1 keeps the square 1, $\exists$ picks one path
 $w \mapsto (1, [5]), (1, [6]), (1, [8]) : A \times L A$ one output per path in w
 $\text{cp}(w) = \{(1, [5]), (1, [6]), (1, [8])\} : E(A + A \times L A)$ the three of them, collected

§10.3.3. $\text{gen} = F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha))$

(10.3.3a)

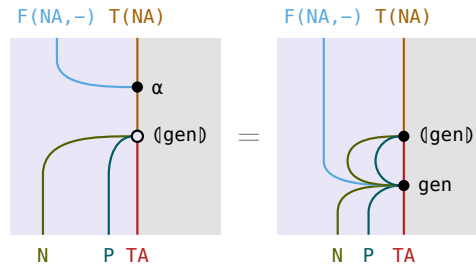


(10.3.3b)

$\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$
 $u = ((1, 2, 3, 4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) : F(N A, N(E(L A)))$
 $F(1, \text{moves trans } N(\text{union}))$ 1 keeps the column, the path SETS move
 $\text{moves}(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$
 $= (\{[6]\}, \{[7]\}, \{[8]\}, \{[5]\}),$
 $(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}),$
 $(\{[8]\}, \{[5]\}, \{[6]\}, \{[7]\})$: $E(N(E(L A)))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= (\{[6]\}, \{[5]\}, \{[8]\}, \{[7]\}, \{[6]\}, \{[5]\}),$
 $\{[8]\}, \{[7]\}, \{[6]\}, \{[5]\}, \{[8]\}, \{[7]\})$: $N(E(E(L A)))$ row k gets rows k-1, k, k+1
 $N(\text{union})(\text{that})$
 $= (\{[5], [6], [8]\}, \{[5], [6], [7]\},$
 $\{[6], [7], [8]\}, \{[5], [7], [8]\})$: $N(E(L A))$ every path into row k, none dropped
 $\text{zip}(\text{that})$
 $= ((1, \{[5], [6], [8]\}), (2, \{[5], [6], [7]\}),$
 $(3, \{[6], [7], [8]\}), (4, \{[5], [7], [8]\}))$: $N(F(A, E(L A)))$ each row: its square, and the paths it may be put in front of
 $N(\text{cp } P(\alpha))(\text{that})$
 $= (\{[1, 5], [1, 6], [1, 8]\}, \{[2, 5], [2, 6], [2, 7]\},$
 $\{[3, 6], [3, 7], [3, 8]\}, \{[4, 5], [4, 7], [4, 8]\})$: $N(E(L A))$ cp pairs the square with each path, α prefixes it: $\alpha(1, [5]) = [1, 5]$

§10.3.4. gen is an F-algebra; $(\text{gen}) : \alpha(\text{gen}) = F(1, (\text{gen})) \text{gen}$

(10.3.4a)

 α puts the column back on the list, then the fold reads all of itthe fold reads the rest under F , then one gen puts the column in front

$$xs = [(1, 2, 3, 4), (5, 6, 7, 8)] = \alpha((1, 2, 3, 4), [(5, 6, 7, 8)]) : L(N A) \quad (10.3.4b)$$

$$(\text{gen})(xs) = \text{gen}((1, 2, 3, 4), (\text{gen})[(5, 6, 7, 8)])$$

both halves are in $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$: the tail folds to $(\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})$, one path per row and each of them one square long, and gen on it is $\text{gen} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp}(F(A, -)) E(\alpha))$'s walk.

§10.3.5. $Q : F(N A, N(L A)) \rightarrow N(L A)$, an $F(N A, -)$ -algebra: $Q = F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$

(10.3.5a)

$$Q : F(N A, N(L A)) \rightarrow N(L A)$$

the fold's algebra: a new column, one path per row

$u = ((1, 2, 3, 4), ([5], [6], [7], [8])) : F(N A, N(L A))$
 $F(1, \text{moves trans } N(\text{est}(R)))$ 1 keeps the column, the paths move
 $\text{moves}([5], [6], [7], [8])$
 $= ([6], [7], [8], [5]),$
 $([5], [6], [7], [8]),$
 $([8], [5], [6], [7])$: $E(N(L A))$ down, unmoved, up
 $\text{trans}(\text{that})$
 $= ([6], [5], [8]), [7], [6], [5],$
 $[8], [7], [6], [5], [8], [7])$: $N(E(L A))$ row k gets rows k-1, k, k+1
 $N(\text{est}(R))(\text{that})$
 $= ([5], [5], [6], [5])$: $N(L A)$ the cheapest into each row – chosen BEFORE the new square
 $\text{zip}(\text{that})$
 $= ((1, [5]), (2, [5]), (3, [6]), (4, [5]))$: $N(F(A, L A))$ each row: its square, and the one predecessor that survived
 $N(\alpha)(\text{that})$
 $= ([1, 5], [2, 5], [3, 6], [4, 5])$: $N(L A)$ $\alpha(1, [5]) = [1, 5]$

$$\text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge F(1, \text{moves trans } N(\text{est}(R))) = [l, \text{pm } r] \wedge \text{zip} = \quad (10.3.5b)$$

$$[N(\text{inl}), \text{zip}' N(\text{inr})] \wedge \text{inl } \alpha = \text{wrap} \wedge \text{inr } \alpha = \text{cons} \Rightarrow Q = [N(\text{wrap}), \text{pm } \text{zip}' N(\text{cons})]$$

at the last column Q starts one path per row, and at each earlier one it puts each square in front of the cheapest of the three kept paths it can step to — the algebra $\text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge$

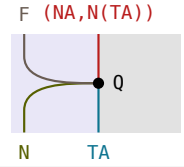
$$\text{lax}(\text{setify}) \wedge R R \sqsubseteq R \wedge F(1, N(\text{est}(R))) Q \sqsubseteq \text{gen } N(\text{est}(R))$$

$$\Rightarrow (Q) \text{ setify } \text{est}(R) \sqsubseteq \text{paths } \text{est}(R) \text{ 's last step folds}$$

circuit — the fork is $F(\text{NA}, N(\text{LA})) = \text{NA} + \text{NA} \times N(\text{LA})$

Hinze–Marsden

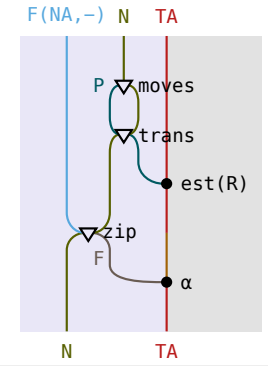
$$F(\text{NA}, N(\text{TA})) \rightarrow \boxed{Q} \rightarrow N(\text{T A})$$



$$F(N(\text{T A}), \text{F}(1, \text{moves trans } N(\text{est}(R)))) \rightarrow F(N(\text{T A}), \text{zip}) \rightarrow N(F(\text{A}, \text{TA})) \rightarrow N(\alpha) \rightarrow N(\text{T A})$$

= the fusion condition $\text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge R R \sqsubseteq R \wedge F(1, \text{est}(R)) \alpha \sqsubseteq \text{cp}(F(\text{A}, -)) E(\alpha) \text{ est}(R)$

$\Rightarrow F(1, N(\text{est}(R))) Q \sqsubseteq \text{gen } N(\text{est}(R))$ read as a definition,



$$F(\text{NA}, N(\text{TA})) \rightarrow \left(\begin{array}{l} \boxed{l} \rightarrow N(p_\alpha) \rightarrow N(\text{wrap}) \\ \boxed{r} \rightarrow q \rightarrow \text{pm} \rightarrow q \rightarrow \text{zip}' \rightarrow N(q_\alpha) \rightarrow N(\text{cons}) \end{array} \right) \rightarrow N(\text{T A})$$

$$\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$$

$$\text{lax(moves)} \wedge \text{lax(trans)} \wedge \text{lax(zip)} \wedge \text{lax(setify)} \wedge R \ R \sqsubseteq R \wedge F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen} \quad (10.3.5c)$$

$$N(\text{est}(R)) \Rightarrow \langle Q \rangle \text{ setify } \text{est}(R) \sqsubseteq \text{paths } \text{est}(R)$$

shortest paths on a cylinder: the dynamic programming approach to exhaustive search allows a path of least cost to be found in $O(n \times m)$ time; Q is $\text{lax(moves)} \wedge \text{lax(trans)} \wedge \text{lax(zip)} \wedge$

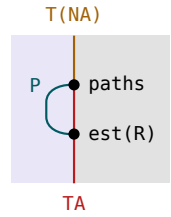
$$F(1, \text{moves trans } N(\text{est}(R))) = [l, pm \ r] \wedge \text{zip} = [N(\text{inl}), \text{zip}' \ N(\text{inr})] \wedge \text{inl } \alpha = \text{wrap} \wedge \text{inr } \alpha = \text{cons}$$

$$\Rightarrow Q = [N(\text{wrap}), pm \ \text{zip}' \ N(\text{cons})] \text{ 's algebra.}$$

circuit

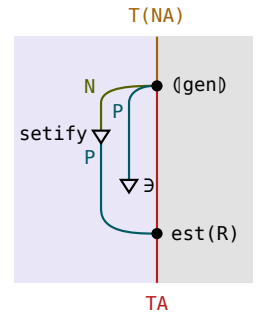
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$$T(N \ A) \text{---} \boxed{\text{paths}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$



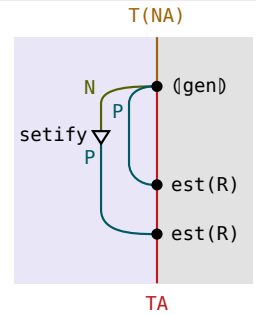
$$= T(N \ A) \text{---} \boxed{F(NA, N(P(TA)))} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{\text{setify}} \text{---} PP(T \ A) \text{---} \boxed{\text{union}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$\text{paths} \triangleq \langle \text{gen} \rangle \text{ setify union at paths}$



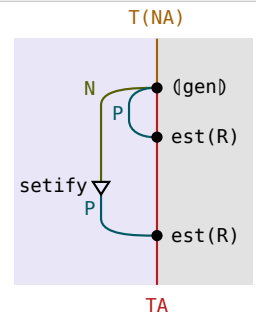
$$\sqsubseteq T(N \ A) \text{---} \boxed{F(NA, N(P(TA)))} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{\text{setify}} \text{---} PP(T \ A) \text{---} \boxed{P(\text{est}(R))} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$P(\text{est}(R)) \text{ est}(R) \sqsubseteq \text{union est}(R)$ — a minimum in each set, then a minimum of those; R transitive



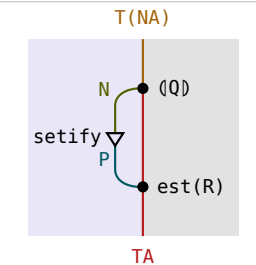
$$\sqsubseteq T(N \ A) \text{---} \boxed{F(NA, N(P(TA)))} \text{---} \boxed{\text{gen}} \text{---} NP(T \ A) \text{---} \boxed{N(\text{est}(R))} \text{---} N(T \ A) \text{---} \boxed{\text{setify}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

$\text{setify lax natural}$



$$\sqsubseteq T(N \ A) \text{---} \boxed{F(NA, N(TA))} \text{---} \boxed{Q} \text{---} N(T \ A) \text{---} \boxed{\text{setify}} \text{---} P(T \ A) \text{---} \boxed{\text{est}(R)} \text{---} T \ A$$

fusion at $\text{lax(moves)} \wedge \text{lax(trans)} \wedge \text{lax(zip)} \wedge R \ R \sqsubseteq R \wedge F(1, \text{est}(R)) \ \alpha \sqsubseteq \text{cp}(F(A, -))$
 $E(\alpha) \text{ est}(R)$
 $\Rightarrow F(1, N(\text{est}(R))) \ Q \sqsubseteq \text{gen } N(\text{est}(R))$



$$\text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge R \subseteq R \wedge F(1, \text{est}(R)) \alpha \subseteq \text{cp}(F(A, -)) E(\alpha) \text{est}(R) \Rightarrow (10.3.5d)$$

$$F(1, N(\text{est}(R))) Q \subseteq \text{gen } N(\text{est}(R))$$

$$\text{fusion: the condition for fusion in } \text{lax}(\text{moves}) \wedge \text{lax}(\text{trans}) \wedge \text{lax}(\text{zip}) \wedge \text{lax}(\text{setify}) \wedge R \subseteq R \wedge F(1, N(\text{est}(R))) Q \subseteq \text{gen } N(\text{est}(R))$$

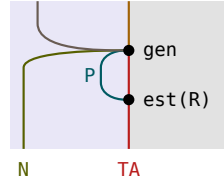
$$\Rightarrow (Q) \text{ setify } \text{est}(R) \subseteq \text{paths } \text{est}(R) \text{ 's last step, used to derive a definition of } Q.$$

circuit

$$F(NA, N(P(TA))) \text{---} \boxed{\text{gen}} \text{---} NP(TA) \text{---} \boxed{N(\text{est}(R))} \text{---} N(TA)$$

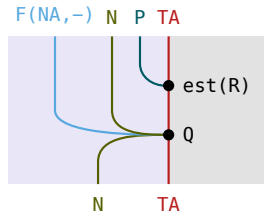
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$$F(NA, N(P(TA)))$$



$$\boxed{\text{FNP}(TA)} \text{---} \boxed{F(1, N(\text{est}(R)))} \text{---} FN(TA) \text{---} \boxed{Q} \text{---} N(TA)$$

(7.13), then zip, trans, moves lax natural



$$(7.13) \text{ is } F(1, \text{est}(R)) \alpha \subseteq \text{cp } P(\alpha) \text{est}(R), \text{ Theorem 7.1 Map}(f) \wedge 1 \subseteq R$$

$$\Rightarrow f \circ F(R) f \subseteq R \Leftrightarrow F(\text{est}(R)) f \subseteq \frac{F(\exists) f}{\exists} \text{est}(R) \text{ at the map } \alpha \text{ with } \frac{F(1, \exists) \alpha}{\exists} = \text{cp } P(\alpha): \text{ extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.}$$